

Supporting Information for: Snowmelt regime and dispersal limitation predict the decadal expansion of dwarf bamboo into alpine snowbed vegetation

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This Supporting Information follows the order of the methods in the main text. Figures (S1–S7) and tables (S1–S12) are numbered in their order of appearance; all are cited from the main text.

Snowmelt regime and trends

Figures S1–S3: temporal and spatial patterns of snowmelt

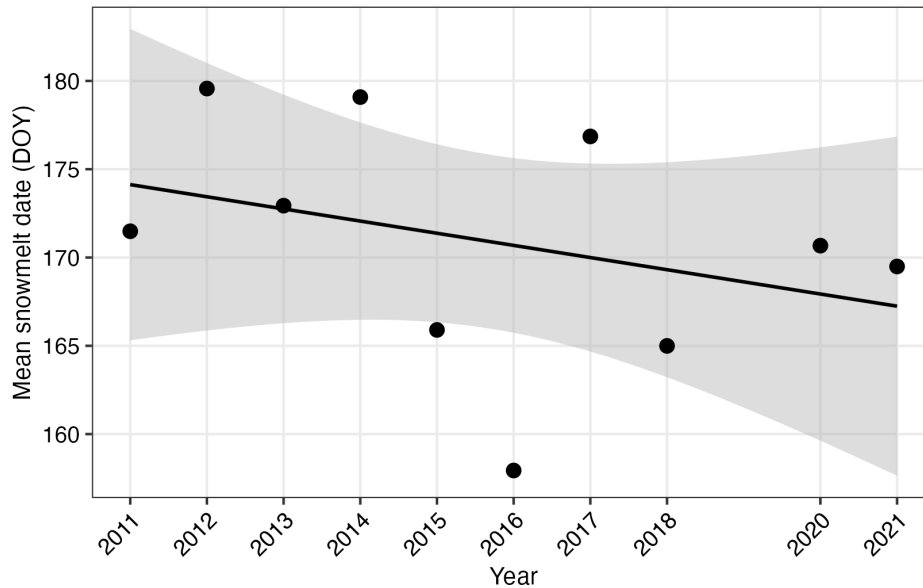


Figure S1: Annual landscape-mean snowmelt DOY varied markedly among years, with no detectable temporal trend. The time series covers 2011–2021, with 2019 missing. The slope of the regression against year was -0.69 days/year (95% confidence interval -2.24 to $+0.87$, $p = 0.34$).

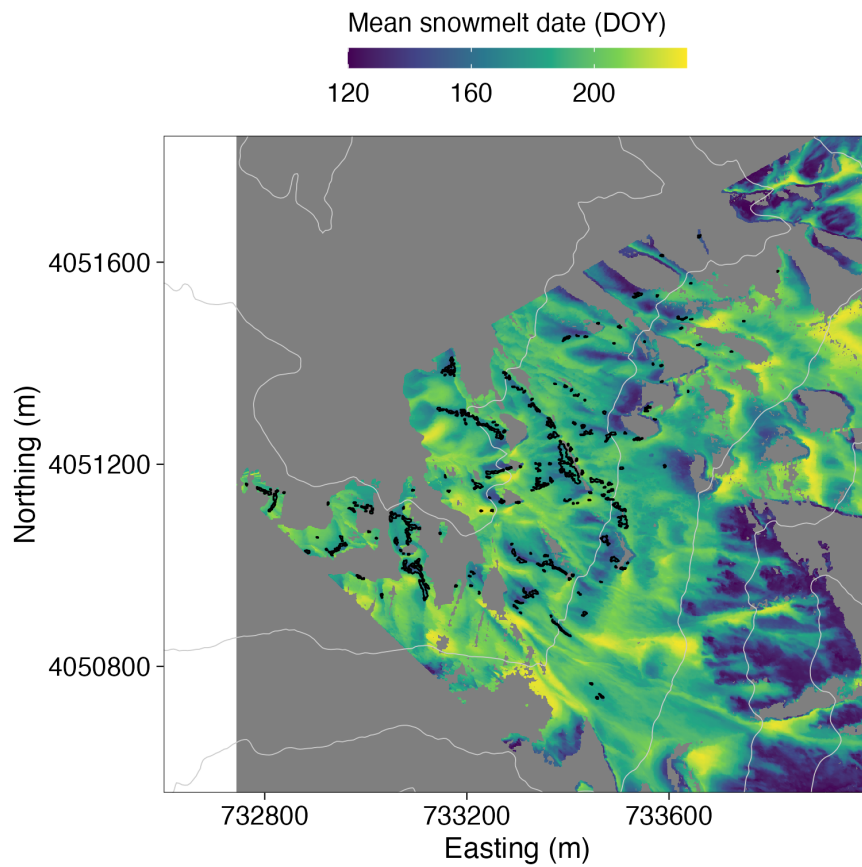


Figure S2: Snowmelt climatology followed the microtopographic gradient from early-melting ridges to late-melting snowbed hollows. The map shows per-pixel mean snowmelt DOY for 2011–2021. Black lines show the 2021 *Sasa* distribution (patches exceeding 5 m²), and grey lines are 100 m contours.

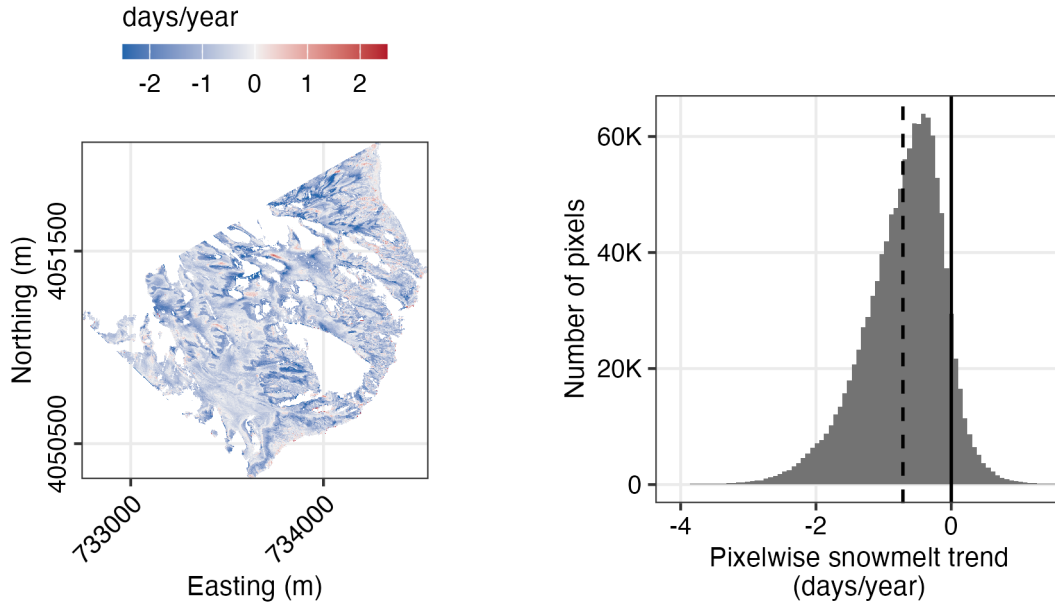


Figure S3: Per-pixel snowmelt trends were negative in 91.7% of pixels, but none remained significant after Benjamini–Hochberg correction. Left: trend map; right: frequency distribution of slopes. The mean slope was -0.71 days/year (median -0.63 , SD 0.63 , $n = 1,206,063$), and only 2.0% of pixels were individually significant ($p < 0.05$). In the right panel, the solid black line denotes zero and the dashed black line denotes the mean (-0.71 days/year). The horizontal axis is truncated at the 0.1–99.9 percentiles; pixels outside this range (0.2% of all pixels, with slopes below -4.1 or above $+1.4$ days/year, including residual-snow margins) are not shown.

Power to detect snowmelt trends

With a 10-year record and a residual annual SD of approximately 9 days, the minimum slope detectable by per-pixel regression at the 5% significance level is approximately 2 days/year. The observed mean slope (-0.71 days/year) is far below this detection limit, so a lack of significance is expected regardless of whether a true trend exists. This is why the main text uses trends only to parameterise scenarios, rather than as evidence of change.

Rationale for excluding trends as predictors: sensitivity analysis adding a trend term to Model B

We added a smooth term for the per-pixel trend, $s(\text{trend})$, to the selected Model B specification and re-evaluated it using the same 4 spatial blocks in leave-one-block-out validation (Table S1).

Table S1: Spatial-block validation of the addition of a snowmelt trend term to Model B. AUC decreased consistently in all 4 blocks.

Evaluation scope	Baseline AUC	+s(trend) AUC	Δ AUC
Block 1	0.9642	0.9636	−0.0006
Block 2	0.9091	0.9084	−0.0007
Block 3	0.9597	0.9595	−0.0002
Block 4	0.9579	0.9574	−0.0005
Pooled	0.9461	0.9456	−0.0005

Although $s(\text{trend})$ was nominally significant in-sample (edf 3.15, $p = 0.011$, $\Delta\text{AIC} -8.1$), it did not improve prediction in held-out blocks and increased explained deviance by only 0.03 percentage points. Across the central 90% of the trend distribution (-1.84 to $+0.12$ days/year), the partial effect spanned only 0.13 on the logit scale (odds ratio 1.13), and the 95% band included zero throughout (Figure S4, right). This was negligible compared with the span of the distance term in the same model (5.8 on the logit scale). Thus, the establishment record could not identify a response to trends, and including the trend as a predictor would only add parameters that the data could not constrain.

Rationale for spatially uniform snowmelt shifts

The scenarios for the 2030 projections apply spatially uniform shifts to snowmelt climatology. We examined the spatial structure of the per-pixel trend field (Figure S4, left). Trends were almost unrelated to elevation (Spearman $\rho = +0.022$; 50 m elevation bins explained 1.9% of the variance), but showed a **non-monotonic** relationship with snowmelt climatology: means across well-sampled 10-day bins ranged from -0.47 days/year in the earliest-melting bin (DOY 120–130), through a minimum of -1.29 (DOY 150–160) and a local maximum of -0.33 (DOY 180–190), to -0.97 in the latest bin (DOY 220–230), and the bins explained 24% of the pixel-level variance. Because the relationship was non-monotonic, Spearman ρ was close to zero (-0.007). A uniform shift is therefore an approximation. Nevertheless, the means of all well-sampled bins were negative and within ± 0.6 days/year of the landscape mean, with deviations smaller than within-bin variation (SD 0.39–0.77 days/year). The scenario range in the main text ($0 / -0.71 / -2.24$ days/year) brackets all bin means. Moreover, as shown above, the trend field did not contribute to predicting establishment. A uniform shift is therefore an adequate and deliberately conservative approximation for the purpose of the scenarios.

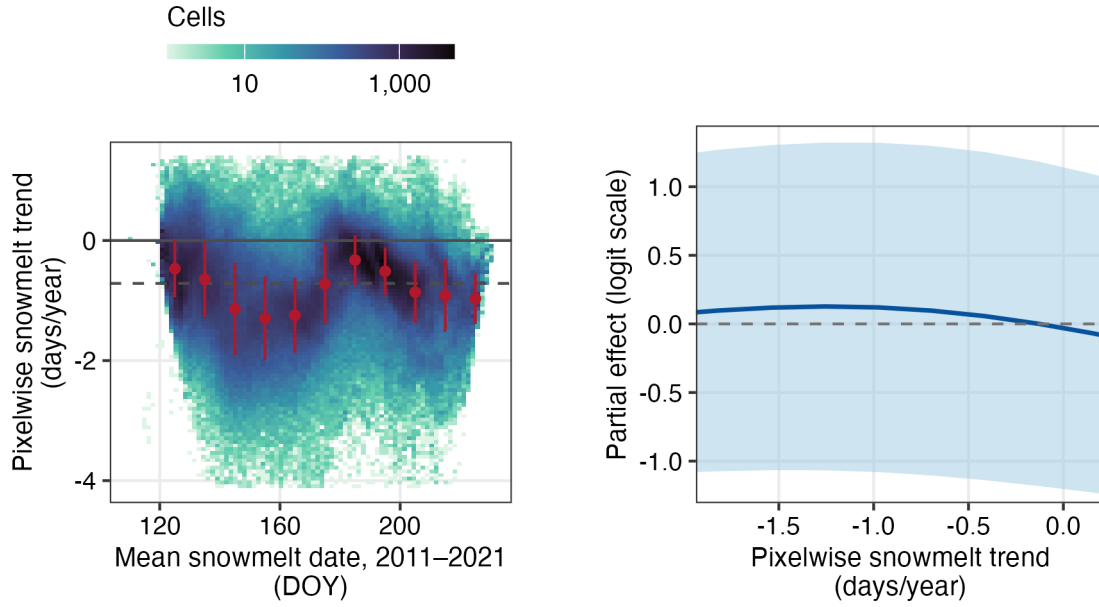


Figure S4: Snowmelt trends varied non-monotonically with snowmelt climatology, but their partial effect in Model B was indistinguishable from zero. Left: two-dimensional binned density of per-pixel trends against snowmelt climatology. Red points show bin means ± 1 SD. The non-monotonic arch was small relative to the vertical scatter. Right: partial effect of $s(\text{trend})$ added to Model B, shown over the central 90% of the trend distribution. The 95% band (grey) included zero throughout.

Vegetation classification

Classifier architecture and accuracy

The classifier was a Conv2D–LSTM operating on 5×5 -pixel patches, using temporal changes in leaf colour across seasonal image sequences as input. The full installation, image-acquisition, and georectification pipeline followed Okamoto, Ide, and Oguma (2024).

Input. Each training or prediction sample was a stack of 7 co-registered seasonal frames covering one 5×5 -pixel neighbourhood, i.e. 7 time steps $\times$ 3 (RGB) channels $\times$ 5 $\times$ 5 pixels. Patches were cut from the aligned image series, resampled to the 5×5 kernel, stored as 8-bit values, and rescaled to the interval $[0, 1]$ before entering the network. The two target years were treated as independent series of 7 frames each, and the same network was trained separately on the 2012 and the 2021 series.

Architecture. Each frame passed through a single convolutional block — Conv2d($3 \rightarrow 8$ channels, 3×3 kernel, stride 1, no padding) $\rightarrow$ PReLU $\rightarrow$ batch normalisation $\rightarrow$ MaxPool2d(3×3 , stride 1) — which reduces a 5×5 patch to a single spatial position and yields 8 features per frame. The resulting 7-step sequence of 8-dimensional feature vectors was passed to a single-layer, unidirectional LSTM with 7 hidden units (the implementation sets the hidden width to the mean of the convolutional output width and the number of classes, i.e. $\lfloor (8 + 7)/2 \rfloor = 7$). The final hidden state was normalised, passed through one fully connected layer of 7 units with PReLU and batch normalisation, and mapped by a second fully connected layer to the 7 class scores, followed by a softmax. Dropout layers were present but were set to a rate of 0, so the fitted network used none.

Training. The loss was cross-entropy and the optimiser RAdam with a fixed learning rate of 1×10^{-3} ; the batch size was 500 and each fold was trained for 50 epochs. No data augmentation was applied: the training and validation transform pipelines are both empty. Training labels comprised 216 hand-drawn polygons, which supplied 618,012 labelled pixels. These were split by stratified 5-fold cross-validation (shuffled, stratified on class), so that each fold trained on 4/5 of the labelled pixels and validated on the remaining 1/5 (mean 123,602 pixels per validation fold). Within each fold the epoch with the lowest validation loss was retained. The published maps are the per-pixel majority vote of the 5 fold models applied to the full image.

Class-specific accuracy is given in Table S2 as the mean across the 5 validation folds. *Sasa*, the focal class, was recovered with an F1 of 0.96 in both years; the weakest class in both years was maple, the rarest of the 7 (12,758 labelled pixels), whose recall fell to 0.90 in 2021. Because the folds were split at the pixel level and the same polygons were reused for both years, these values describe accuracy near the training polygons and are upper bounds on accuracy elsewhere in the scene.

Table S2: Class-specific precision (P), recall (R), and F1 of the Conv2D–LSTM classifier, as means over the 5 stratified cross-validation folds. “Pixels” is the total number of labelled pixels of the class across all folds; the same label polygons were used for both years. The weighted averages are the overall accuracies quoted in the Methods (0.982 for 2012, 0.975 for 2021).

Class	Pixels	2012 P	2012 R	2012 F1	2021 P	2021 R	2021 F1
<i>Sasa</i>	38,396	0.967	0.953	0.960	0.958	0.952	0.955
Other vegetation	339,439	0.988	0.993	0.990	0.987	0.989	0.988
No vegetation	40,632	0.979	0.975	0.977	0.968	0.974	0.971
Rowans	29,282	0.972	0.976	0.974	0.945	0.938	0.941
Maple	12,758	0.971	0.935	0.953	0.947	0.896	0.921
Montane alder	48,130	0.984	0.970	0.977	0.963	0.947	0.955
Dwarf pine	109,375	0.972	0.972	0.972	0.963	0.972	0.968
Macro average	618,012	0.976	0.968	0.972	0.962	0.953	0.957
Weighted average	618,012	0.982	0.982	0.982	0.975	0.975	0.975

Figure S5: vegetation classification maps for 2012 and 2021

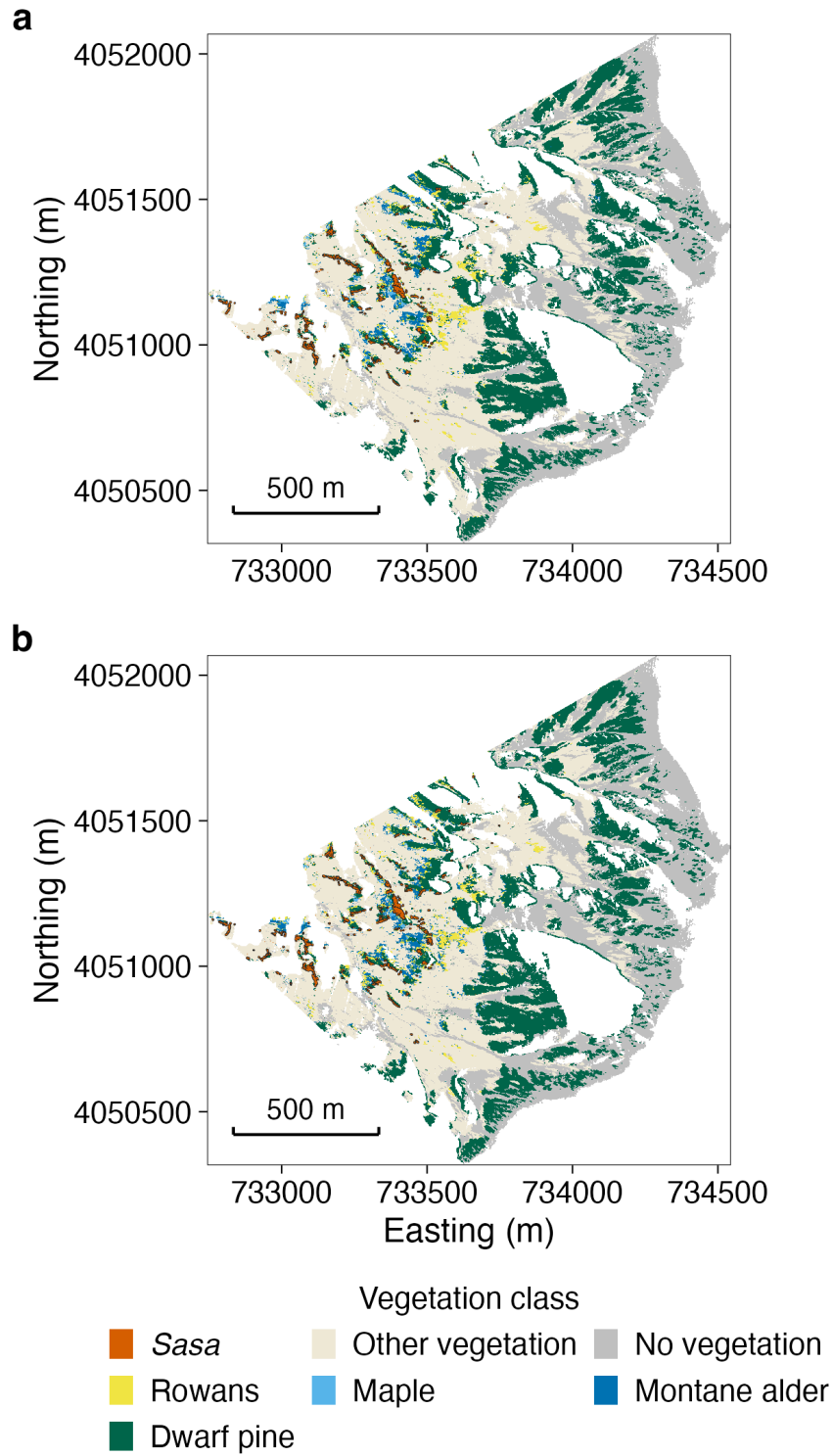


Figure S5: The 7-class vegetation classification⁸ was stable between the two years, and change was concentrated at the margins of the *Sasa* patches. (a) 2012 and (b) 2021, on the 1 m analysis grid over the full analysis extent. Both panels share the legend below. *Sasa* covered 8,547 cells in 2012 and 10,174 cells in 2021 (0.7% and 0.8% of the classified area); the shares of the other six classes changed by at most 3 percentage points.

Table S3: full transition matrix for all classes (2012→2021)

Rows indicate the 2012 class and columns the 2021 class. Values are cell counts (1 cell $\approx$ 1 m²).

Table S3: Full transition matrix for all vegetation classes, 2012→2021 (cell counts).

2012 \ 2021	<i>Sasa</i>	Other vegetation	No vegetation	Rowans	Maple	Montane alder	Dwarf pine	Total
<i>Sasa</i>	6,079	932	3	159	85	133	1,156	8,547
Other vegetation	2,170	432,162	32,001	3,031	380	7,187	9,385	486,316
No vegetation	13	5,096	342,186	49	0	16	10,569	357,929
Rowans	224	3,734	30	17,282	871	2,682	2,104	26,927
Maple	155	624	0	1,271	2,523	703	424	5,700
Montane alder	109	3,643	45	2,023	234	13,033	1,683	20,770
Dwarf pine	1,424	5,488	6,418	1,369	209	1,189	278,650	294,747

Spatial verification of *Sasa*→dwarf pine transitions: adjacency analysis

To assess whether *Sasa*→dwarf pine transitions (1,156 cells) were concentrated at the boundaries of existing dwarf pine, we calculated the Euclidean distance from each transition cell to the nearest 2012 dwarf pine cell (Table S4).

Table S4: Distribution of distances to 2012 dwarf pine. Of the cells transitioning to dwarf pine, 87.5% were directly adjacent (median distance 1.0 m), approximately 1.9 times the proportion in the reference distribution (46.6%).

Group	n	≤ 1.5 m (directly adjacent)	1.5–3 m	3–5 m	>5 m
<i>Sasa</i> →dwarf pine	1,156	87.5%	8.8%	2.0%	1.6%
<i>Sasa</i> →other vegetation and other classes	1,312	39.6%	23.6%	16.9%	19.9%
All 2012 <i>Sasa</i> cells (reference)	8,547	46.6%	27.6%	15.2%	10.7%

The transition rate to dwarf pine in the adjacent band (≤ 1.5 m) was 25.4%, approximately 6 times that in the 1.5–3 m band (4.3%), and levelled off at around 2% beyond 3 m, consistent with a classification-noise floor. Isolated transitions more than 5 m away amounted to only 19 m² (1.6% of transition area; 10 clusters, maximum 4 m²). Of these, only 2 clusters totalling 5

m², located 21–24 m from the front, were difficult to explain biologically and were consistent with sporadic misclassification within *Sasa* stands.

Blind interpretation of paired images

We conducted a blind interpretation of paired images to assess whether lateral growth of dwarf pine branches could be distinguished from shifts in the classification of mixed boundary pixels. The 129 samples comprised 79 stratified samples of loss cells (37 to dwarf pine, 30 to other vegetation, and 12 to deciduous shrubs, proportional to the population composition of destination classes, 47/38/15%), 40 stable controls (20 *Sasa* in both years and 20 dwarf pine in both years), and 10 reverse transitions (dwarf pine→*Sasa*). Paired image crops from 2012 and 2021 were presented in random order with transition types concealed. Blinding was to transition type only: the interpreter approached the samples as a set of mapped loss cells and was not informed that stable controls and reverse transitions had been interleaved. Because this expectation was identical across strata, it cannot explain the agreement between loss cells and controls reported below. Each sample contained 2 pairs, one from the green-leaf season and one from the autumn-colour season; frames with similar image greenness were paired to reduce phenological differences, and a box marked the projected location of the target 1 m cell. The interpreter (the first author) assigned each sample to one of 4 interpretation categories: *Sasa* loss, canopy cover, boundary ambiguity, or indeterminate. Residual georectification errors could shift the box by 1–2 cells. Boundary ambiguity was used when no clear change was visually discernible.

Table S5: Distribution of blind interpretation judgements by sample group (*percentages may not sum to 100% because of rounding).

Sample group	n	Boundary ambiguity	Canopy cover	<i>Sasa</i> loss	Indeterminate
<i>Sasa</i> →dwarf pine	37	73%	22%	3%	3%
<i>Sasa</i> →other vegetation	30	87%	7%	3%	3%
<i>Sasa</i> →3 deciduous shrub classes	12	75%	8%	8%	8%*
Stable control (<i>Sasa</i> → <i>Sasa</i>)	20	85%	10%	5%	0%
Stable control (dwarf pine→dwarf pine)	20	80%	15%	5%	0%
Dwarf pine→ <i>Sasa</i>	10	80%	10%	0%	10%

Judgement distributions were almost identical for loss cells and stable controls. Thus, the interpreter could not distinguish cells mapped as changed from those mapped as unchanged using the photographs. Canopy cover was assigned to 22% (8/37) of *Sasa*→dwarf pine transitions, only slightly above the corresponding judgement for stable dwarf pine controls (15%, 3/20, providing an indication of the false-positive rate). **At the photographic scale, the nature**

of transitions in individual 1 m cells could not be resolved: whether they represented lateral growth or shifts in classification, or whether *Sasa* survived beneath the canopy. The main text’s interpretation of apparent loss to dwarf pine as a boundary phenomenon therefore rests on the concentration of adjacent transitions (preceding section) and the physical constraint imposed by dwarf pine annual shoot extension of approximately 4–5 cm/year (Amagai, Kaneko, and Kudo 2015), rather than on this interpretation exercise. All 129 samples were tallied, including 5 that fell below the pre-specified image-quality screen; excluding them does not change the pattern. We retain the conservative treatment of loss area as an upper bound on true decline. Raw interpretation data and reproduction instructions are included in `review/image_interpretation_package/` in the repository.

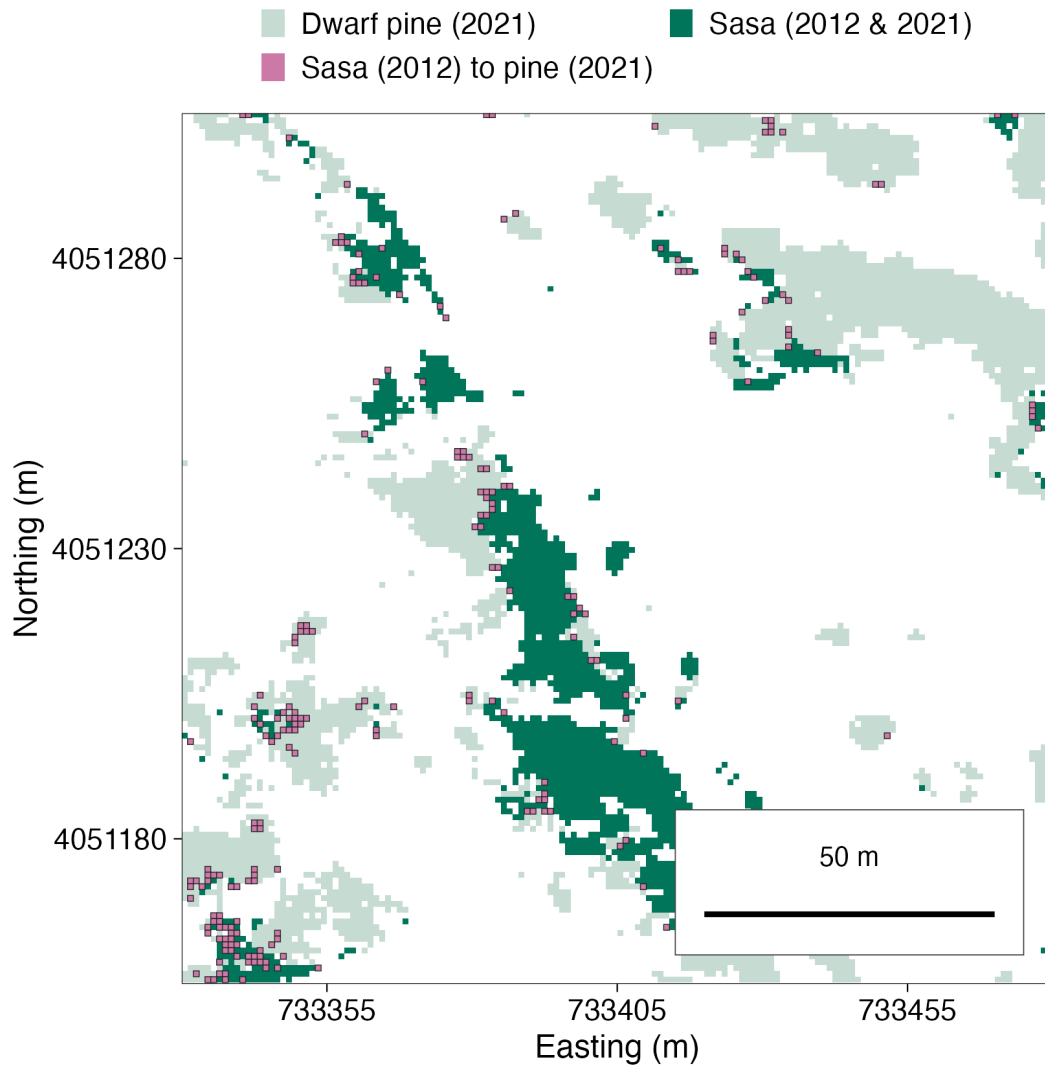


Figure S6: Cells transitioning from *Sasa* to dwarf pine were distributed along the boundaries between persistent *Sasa* and 2021 dwarf pine. This example shows a 150 m window with dense *Sasa*–dwarf pine boundaries. Transition cells are magenta, persistent *Sasa* is dark green, and 2021 dwarf pine is pale green. The window contains 218 transition cells.

Model A

Table S6: held-out performance and ensemble composition

Table S6: Held-out performance of Model A for the full and restricted elevation ranges.

Variant	Evaluation scope	AUC	TSS	TSS-maximising threshold
Main model (full elevation range)	Block mean	0.949	0.785	—
Main model (full elevation range)	Pooled	0.944	0.765	0.158
Elevation-restricted (<2,560 m)	Block mean	0.857	0.583	—
Elevation-restricted (<2,560 m)	Pooled	0.846	0.553	0.302

The lower performance of the elevation-restricted variant (TSS 0.78→0.58) reflects the ease of discriminating high-elevation cells where *Sasa* cannot occur, rather than deterioration of the model. LASSO stacking retained 4 members in the main model (2 gradient boosted tree models, 1 MaxEnt model, and 1 random forest). Boosted trees and MaxEnt accounted for approximately 97% of the weight, and no GAM members remained.

Terrain predictors and Table S7: predictor correlation matrix and VIF

Elevation was taken from the Fundamental Geospatial Data DEM of the Geospatial Information Authority of Japan (Geospatial Information Authority of Japan 2026), whose native 0.2×0.2 grid corresponds to cells of 4.97 m (east–west) $\times$ 6.16 m (north–south) after projection to JGD2000 / UTM zone 53N (EPSG:3099). Slope, aspect, and TPI were derived from this DEM with the GDAL `gdaldem` utility: slope and aspect with the Zevenbergen–Thorne four-neighbour algorithm (Zevenbergen and Thorne 1987), and TPI as the difference between the elevation of the focal cell and the mean elevation of its eight neighbours (a fixed 3×3 window). We verified this provenance post hoc by re-deriving each layer from the archived DEM with `gdaldem` (GDAL 3.12), which reproduced the archived rasters exactly (maximum absolute difference 0). TWI (Beven and Kirkby 1979) was computed in QGIS with SAGA GIS (Conrad et al. 2015) at the tool’s default settings; the specific module, its parameters, and the QGIS and SAGA version numbers were not recorded at the time of processing and could not be recovered, although the archived layer is consistent with a multiple-flow-direction contributing area combined with a lower bound on slope. Aspect was decomposed into northness and eastness at the native DEM resolution, and all terrain layers were then reprojected onto the 1 m analysis grid by

bilinear interpolation; their effective resolution therefore remains that of the source DEM rather than 1 m. The derived terrain rasters are included in the archived deposit. The full predictor correlation structure is given in Table S7.

Table S7: Pairwise Spearman correlations and variance inflation factors (VIF) of the retained predictors. The largest $|r|$ is 0.63 between TWI and snowmelt climatology, and the largest VIF is 2.30 (snowmelt climatology); both are below conventional thresholds ($|r| < 0.7$, $VIF < 5$).

	slope	elevation	TPI	TWI	northness	eastness	snowmelt clim.	VIF
slope	1	0.55	0.00	−0.46	−0.29	−0.14	−0.55	1.75
elevation	0.55	1	0.04	−0.34	−0.19	−0.25	−0.58	1.85
TPI	0.00	0.04	1	−0.44	0.01	0.05	−0.33	1.39
TWI	−0.46	−0.34	−0.44	1	0.19	0.02	0.63	1.92
northness	−0.29	−0.19	0.01	0.19	1	0.48	0.27	1.23
eastness	−0.14	−0.25	0.05	0.02	0.48	1	0.13	1.22
snowmelt clim.	−0.55	−0.58	−0.33	0.63	0.27	0.13	1	2.30

Table S8: threshold sensitivity of suitable area in the main model

Table S8: Sensitivity of suitable area to the binarisation threshold in the main Model A.

Threshold	Suitable area (10^3 m^2)	Ratio to occupied area ($10.2 \times 10^3 \text{ m}^2$)
0.10	272.1	26.7
0.16 (maximum TSS)	164.1	16.1
0.20	136.4	13.4
0.30	89.2	8.8
0.40	52.2	5.1
0.50	23.5	2.3
0.55	13.2	1.3
0.60	5.5	0.5
0.70	0.0	0.0

Suitable area exceeded occupied area at thresholds up to 0.55. At thresholds of 0.60 or above, it fell below occupied area. The statement that suitable area exceeded occupied area therefore uses

the TSS-maximising threshold as its reference (0.16; ratio 16.1), and the main text explicitly states that the relationship held up to a threshold of 0.55.

Figure S7: seed-dispersal stratum (environmental potential)

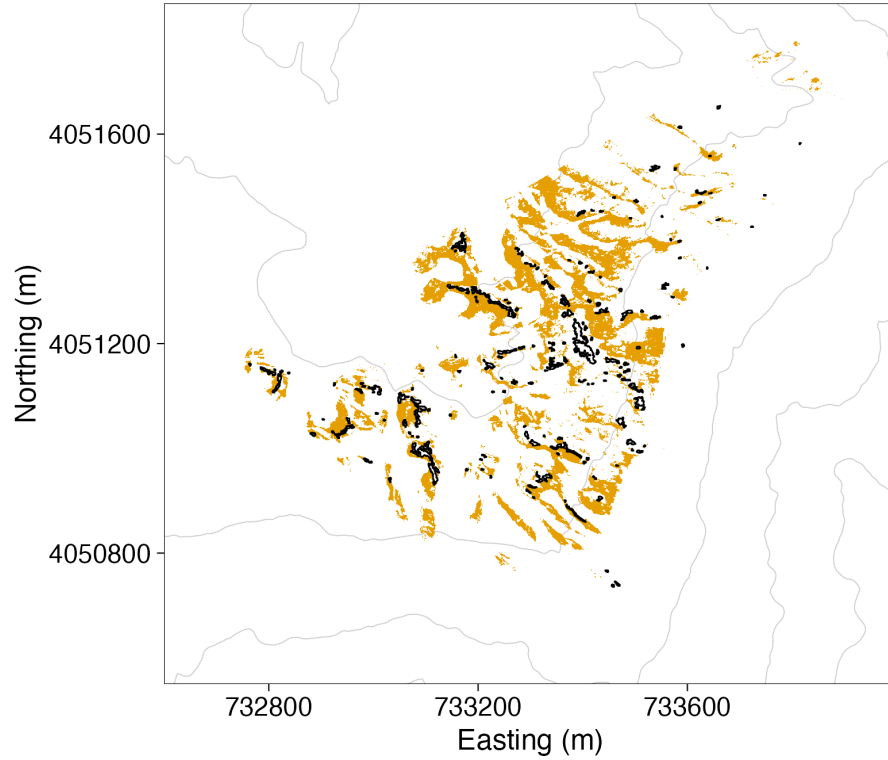


Figure S7: The seed-dispersal stratum was more than an order of magnitude larger than the expected establishment area in the near-term rhizome stratum. Orange cells were classified as other vegetation in 2021 and had Model A suitability at or above the TSS-maximising threshold (0.158), covering a nominal area of $84.7 \times 10^3 \text{ m}^2$ compared with expected establishment of 2,429–3,383 m^2 in the near-term rhizome stratum. Black lines show the 2021 *Sasa* distribution.

Model B

Selection of the distance-term specification

We selected the distance-term specification by comparing a smooth on the linear distance axis, $s(d)$, with a smooth on log-transformed distance, $s(\log(1+d))$, using 3 criteria fixed in advance (Table S9).

Table S9: Comparison of distance-term specifications using predefined criteria (linear axis $\rightarrow$ log distance). All 3 criteria were met, so $s(\log(1+d))$ was selected.

Criterion	Requirement	Result	Decision
Deterioration in pooled AUC	≤ 0.005	-0.0096 (improvement)	Pass
Reduction in absolute error in the 0–5 m band	$\geq 30\%$	87.3% reduction	Pass
Deterioration in the 5–20 m band	None	Improvement	Pass

The selected specification achieved block-specific AUC of 0.91–0.96 (pooled AUC 0.946, TSS 0.75). The gradient boosted tree ensemble used as a robustness comparison, fitted on the same folds to all establishment cells and a fold-stratified random sample of 150,000 non-establishment cells, achieved a pooled AUC of 0.948, indicating that the additive model captured nearly all the signal available to a flexible learner. MaxEnt was excluded from the comparison because quasi-separation of the classes along the distance gradient prevented convergence of the regularisation path in `maxnet`/`glmnet`. The selected specification used a basis dimension of $k = 5$ for $s(\log(1 + d))$ and $k = 10$ for the environmental smooths; the distance term is deliberately low-dimensional because establishment records are sparse and larger bases produced unsupported oscillations between distant stands. Kernels fitted with $k = 3, 5$, and 10 on the linear distance axis are included in the deposited output (`model_B_distance_kernel.csv`).

Cellular-automaton projections

Exactness of the decomposition and random-number control

The environmental component of the linear predictor was precomputed for each scenario $\times$ year, and only the distance term was updated within the simulation. The maximum reconstruction error of this decomposition was $|\Delta p_9| < 8.3 \times 10^{-11}$ (checked on 10,000 cells; tolerance 1×10^{-6}), so the decomposition can be regarded as exact. Each replicate was assigned a fixed random seed, independently of parallel scheduling.

Validation and structural sensitivity

The hindcast initialised from the 2012 distribution reproduced observed gross expansion with a ratio of 1.077 (expected 4,414 m² vs observed 4,093 m²), and cell-level AUC was 0.92. In spatial-block-exclusion validation (50 replicates per block), AUC ranged from 0.79 to 0.86 and block-level ratios of expected to observed area ranged from 0.67 to 2.0. Structural sensitivity (50 replicates per setting) is summarised in Table S10.

Table S10: Structural sensitivity of projected establishment area.

Sensitivity setting	Expected area in 2030 ($s = -0.71$)	Ratio to baseline
Baseline	5,730 m ²	1.00
Remove the 5 m ² patch rule	17,226 m ²	3.01
Replace annualisation with the first-order approximation $p_9/9$	5,326 m ²	0.93

Table S11: projection summary by scenario

Table S11: Expected new establishment area by 2030 under each snowmelt scenario.

Scenario	Expected new establishment area by 2030	SD across replicates	Number of replicates
$s = 0$ (no change)	4,901 m ²	70	200
$s = -0.71$ days/year (point estimate)	5,730 m ²	84	200
$s = -2.24$ days/year (lower CI limit)	5,524 m ²	81	200

The extreme scenario (-2.24 days/year) yielded less establishment than the point-estimate scenario. At sites immediately adjacent to the front, where establishment was concentrated, excessive snowmelt advance shifted conditions beyond the favourable snowmelt window (see the Discussion of the main text).

Composition of communities at risk

Table S12: expected establishment area by community (Ministry of the Environment 1:25,000 vegetation map)

Expected establishment area by 2030 is broken down by community within cells classified as other vegetation in 2021. For comparison, the composition of observed 2012–2021 establishment originating from other vegetation is also shown.

Table S12: Expected establishment area by mapped community and scenario, with observed 2012–2021 establishment for comparison.

Community	$s = 0$	$s = -0.71$	$s = -2.24$	Observed 2012–2021
<i>Fauria crista-galli</i> – <i>Carex blepharicarpa</i> community	1,774 m ² (73.0%)	2,500 m ² (73.9%)	3,013 m ² (74.7%)	2,736 m ² (66.8%)
<i>Anaphalis alpicola</i> – <i>Phyllodoce aleutica</i> community	383 m ² (15.8%)	501 m ² (14.8%)	567 m ² (14.1%)	467 m ² (11.4%)
<i>Vaccinium vitis-idaea</i> – <i>Pinus pumila</i> community	251 m ² (10.3%)	346 m ² (10.2%)	393 m ² (9.7%)	882 m ² (21.5%)
Other map categories	21 m ² (0.9%)	36 m ² (1.1%)	59 m ² (1.5%)	11 m ² (0.3%)

The allocation to the *Vaccinium vitis-idaea*–*Pinus pumila* community reflects the scale mismatch between 1:25,000 polygons and the 1 m grid (herbaceous cells within the polygons), rather than establishment beneath dwarf pine canopies.

References

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